

KPC Revenue Assurance: Synthetic Data Dictionary

This document outlines the data architecture and granular schema definitions for the KPC Revenue Assurance (Order-to-Cash & Data Quality) project. The datasets represent an end-to-end operational flow, deliberately injected with real-world anomalies, shrinkages, and data quality flaws to simulate revenue leakage.

I. Data Architecture & Reconciliation Flow

1. The Master Data (Ground Truth)

- **products & depots**: The foundational reference tables for authorized KPC facilities and official EPRA fuel codes.
- **omcs**: The master list of clients (Oil Marketing Companies). Without this, identifying the responsible party for leaked revenue is impossible. It serves as the central hub connecting physical dispatches to financial payments.
- **tariffs**: Holds dynamic pricing rules for pipeline transport and storage. Separating tariffs from transactional data allows for accurate recalculations of expected revenue based on specific effective dates (SCD Type 2), rather than hardcoded assumptions.

2. The Physical vs. Commercial Core (Finding Unbilled Revenue)

- **depot_loading_logs**: Represents the physical reality—what the metering system recorded at the loading bay before the truck drove off.
- **dispatches**: Represents the commercial reality—the official waybill that KPC intends to bill for.
- *Relevance*: Reconciling loading logs against dispatches uncovers "unmetered loading" (product that left the depot but never received a commercial waybill).

3. The Financial Flow (Finding Uncollected Revenue)

- **invoices**: What KPC officially billed the OMCs.
- **payments**: The actual cash KPC received in the bank.
- *Relevance*: Reconciling dispatches against invoices identifies "unbilled leakage" (forgotten, uninvoiced dispatches). Reconciling invoices against payments highlights underpayments, split installments, or entirely unpaid invoices.

4. Operational & Technical Governance

- **depot_daily_inventory**: Provides daily snapshots of physical tank dips versus book balances. Allows for the detection of macro-level product shrinkage or physical leakage before the product even reaches the loading gantry.
- **quarantine_audit_log**: The ETL quality shield. Instead of silently dropping bad data (e.g., missing dates, duplicate records, string formatting errors), the pipeline safely quarantines them here for audit and debugging purposes.

II. Dataset Schemas

1. products.csv (Product Master)

Column Name	Description
product_id	The unique identifier/code for the petroleum product (e.g., PMS, AGO, DPK).
product_name	The official descriptive name of the product.
unit_price_kes	Illustrative unit price placeholder in local currency (where applicable).
is_active	Status flag indicating if the product code is currently active.

2. depots.csv (Depot Master)

Column Name	Description
depot_id	The unique natural key identifier for the physical depot (e.g., Nairobi, Kisumu).
depot_name	The formal name of the depot facility.
location	Geographic location mapping (null in the baseline dataset).
capacity_litres	The total physical storage capacity of the depot in liters.
is_active	Status flag determining if the depot is currently operational.

3. omcs.csv (The Client Registry)

Column Name	Description
omc_id	The primary key for identifying each Oil Marketing Company.
customer_name	The formal commercial title under which the OMC is registered.
kra_pin	The mandatory tax identification number for regulatory compliance.
payment_terms_days	The contractual window provided for settling outstanding invoices.
credit_limit_kes	The financial ceiling for credit exposure in local currency.
risk_rating	A categorical assessment of the client's financial stability (Low/Medium/High).
risk_profile	The assigned behavior tier used to generate specific leakage/anomaly patterns (Good, Small, Medium, High).
contact_email	The designated digital channel for all billing-related correspondence.
phone	The primary telephonic link for operational queries.

is_active	A status flag determining if the client is currently authorized for loading.
------------------	--

4. **tariffs.csv** (The Pricing Engine)

Column Name	Description
tariff_id	The unique version control identifier for specific rate sets.
product	The specific fuel grade or type governed by these rates.
depot	The specific physical KPC facility where these tariffs are enforced.
pipeline_tariff_per_m3_km	The unit cost for transportation via the pipeline network.
storage_tariff_per_m3_day	The daily holding cost per unit volume within the tanks.
effective_start	The activation date for this pricing version.
effective_end	The expiration timestamp for these specific financial rules.

5. **depot_loading_logs.csv** (The Physical Meter Truth)

Column Name	Description
loading_id	The unique digital footprint of a physical loading event.
gantry_bay_id	The specific hardware location where the transfer occurred.
meter_id	The identifier for the flow meter responsible for volume capture.
dispatch_id	The link to the commercial waybill; an empty/null value here flags potential unbilled leakage.
omc_id	The client entity receiving the physical product.
depot	The geographic site of the gantry infrastructure.
product	The classification of the petroleum being dispensed.
physical_loaded_liters	The raw volume recorded by the meter during the event.
temperature_c	The thermal state of the fuel, critical for volume correction calculations.
density_kg_m3	The measured density, used to validate product quality and mass.

loading_timestamp	The precise moment the physical loading was completed.
--------------------------	--

6. **dispatches.csv** (Commercial Intent)

Column Name	Description
dispatch_id	The official tracking number for the commercial waybill.
loading_id	The technical bridge back to the physical meter readings.
date	The calendar day the dispatch was officially entered into the books.
year	A pre-calculated field for annual trend analysis.
month	A temporal dimension for monthly financial reporting.
omc_id	The OMC assigned as the responsible party for the dispatch.
customer_name	The descriptive name of the purchasing entity.
product	The fuel variety specified on the commercial waybill.
depot	The originating KPC facility for this commercial transaction.

volume_liters	The volume intended for billing (to be reconciled against physical logs).
distance_km	The journey length used to calculate transportation fees.
transport_tariff_kes	The computed transport revenue based on current tariffs.
storage_tariff_kes	The calculated storage revenue owed for depot usage.
value_kes	The total commercial worth of the dispatch (Sum of transport + storage fees).

7. **invoices.csv** (The Billing Record)

Column Name	Description
invoice_id	The primary identifier for the financial demand note.
dispatch_id	The specific commercial event being liquidated by this invoice.
omc_id	The OMC debtor identified in the financial system.
customer_name	The entity name appearing on the official invoice document.
product	The product line being charged to the client.
date	The formal issuance date of the financial claim.

value_kes	The total monetary amount the client is expected to pay.
------------------	--

8. **payments.csv** (The Cash Reality)

Column Name	Description
payment_id	The unique receipt number for funds landing in the bank.
invoice_id	The specific bill this payment is intended to clear.
omc_id	The client entity that initiated the transfer.
customer_name	The OMC identified as the source of payment.
date	The timestamp of when the cash was recognized in KPC accounts.
value_kes	The actual liquid amount received from the OMC.
installment_no	Tracking for partial settlements (e.g., payment 1 of 3).

9. **depot_daily_inventory.csv** (Macro Reconciliation)

Column Name	Description
record_id	The unique reference for the daily inventory audit.

date	The specific 24-hour window for the stock take.
depot	The physical storage terminal being audited.
product	The fuel type under macro-reconciliation.
opening_stock_liters	The inventory volume at the start of the operational day.
received_pipeline_liters	Fresh product volume introduced via the pipeline network.
dispatched_liters	The total volume that exited via the loading gantry.
book_closing_stock_liters	The expected inventory based on paper transactions.
physical_dip_closing_stock_liters	The actual volume measured via physical tank dipping.
variance_liters	The critical delta between book and physical stock (flags physical shrinkage or recording errors).

10. quarantine_audit_log (The Quality Shield)

Column Name	Description

quarantine_id	The unique marker for a data row that failed validation.
dataset_name	The upstream table/file that produced the faulty record.
failed_gate	The specific ETL rule that triggered the isolation (e.g., Duplicate Check, Null Date).
reason	A detailed explanation of the data quality failure.
quarantined_at	The precise moment the record was intercepted by the pipeline.
raw_record_json	The original corrupted data preserved for audit and downstream debugging.